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# GORGO: Online Tuning for Cross-Region Network-Aware LLM Serving

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## Abstract

Increasingly, LLM inference services proxy client requests to engine replicas distributed globally. Load-balancing policies must jointly account for factors including KV-cache locality, replica load, and variable network latency when optimizing for metrics like latency and TTFT. However, existing systems only evaluate a subset of these factors in their cost model, leading to uneven concentrations of load and KV-cache across replicas. We present GORGO, a proxy architecture that holistically factors network latency, prefill cost, and queueing delay using tunable parameters. Since open-source chat datasets such as LMSYS-Chat-1M and WildChat-4.8M lack long-context, high prefix-reuse data, we release a synthetic dataset, ART-Chat-2.5M, from long-context production metadata. On a tuning window from ART-Chat-2.5M, evolutionary strategies guide the GORGO policy’s parameters to directly optimize p95 TTFT. During held-out evaluation windows, we fix the parameter values learned from tuning and improve p95 TTFT by 6.9–15.5% and p95 end-to-end (E2E) latency by 14.3–30.9% over baseline load-balancing policies such as simple session affinity and prefix-cache. The code and ART-Chat-2.5M dataset can be found at <https://github.com/atoniolo76/GORGO>.

## 1 Introduction

In LLM serving systems, perceived latency to the user is dominated by the time-to-first-token (TTFT). On a single replica, TTFT is dominated by three costs: (i) prefill time, (ii) round trip time (RTT) from client (proxy) to replica, and (iii) queueing delay behind in-flight requests. Prefix-caching, which is enabled in inference engines SGLang [Zheng et al., 2024b] and vLLM [Kwon et al., 2023], eliminates the prefill cost of previous turns in a multi-turn conversation. As LLM context windows increase in length, the time saved by prefix caching 90% of a prompt with 100,000 tokens reduces the prefill cost to 10,000 tokens and decreases TTFT substantially.

Since LLM deployments proxy requests to inference engines across regions, the cost savings of prefix-caching depend on choosing a replica with the request session’s prefix. Popular routing policies such as consistent hashing and prefix reuse aim to distribute load evenly while creating affinity between a session’s requests and replica(s) to maximize KV-cache reuse [Karger et al., 1997, Stoica et al., 2003, Zheng et al., 2024b, Xia et al., 2025]. In compute-constrained regimes, bursty workloads can saturate a high-affinity replica, causing head-of-line (HOL) queueing delays from decode memory contention and negating cost savings from prefix-cache reuse. Routing policies that holistically evaluate all costs related to TTFT can maintain high prefix-cache reuse while minimizing the negative effects of load saturation and heterogeneous network latency.

Existing load balancing policies such as least-load, session affinity, and prefix-reuse may account for replica load or KV-cache hit rate; however, no existing policies consider network latency in cross-region scenarios, which can range on the order of 10ms to 1s (Figure 2). GORGO’s routing policy accounts for all three TTFT costs, normalizes the units of measurement via tunable parameters, and jointly optimizes parameters through online tuning on real user workloads. To tune and stress-test different routing policies, in (§3) we compile an LLM traffic trace from real production requests with high prefix-reuse and long-context prompts. The trace follows Mooncake’s FAST’25 format [Qin et al., 2025] containing per-request timestamps, which can be linearly scaled to simulate variable saturation profiles.

We benchmark GORGO on a series of user workloads from ART-Chat-2.5M, our sensitized production Mooncake trace, and sweep across variable time scales to effectively saturate replicas without simulating unrealistic HOL queueing delay. Over existing load balancing policy baselines, GORGO jointly balances optimal TTFT with request concentration across replicas. Under the continuous batching paradigm, the ES-driven hillclimb tuner exploits warmer replicas close to the proxy and dramatically reduces TTFT at the cost of end-to-end (E2E) latency. We characterize the trade-off between request concentration across replicas, which inflates E2E latency for unbalanced distributions, and TTFT. Our contributions help contextualize the performance of LLM proxy routing policies in real-world user workloads, and (§5) lists a case-by-case scenario of when one would want to use aforementioned baseline policies, the online GORGO policy, and offline GORGO with held-out weights. Finally, we show how conditioning the GORGO cost model on proxy-recorded replica load mitigates subversion of TTFT via continuous batching and slashes request latency across a panoply of metrics.

## 2 GORGO

### 2.1 Design Space

For LLM load balancing policies, the design space includes which routing signals to consider and methods for choosing environment settings.

**Routing Signals:** Across existing load balancing policies, two distinct routing signals are considered: (i) replica load and (ii) KV-cache locality. Simple policies such as least-load and least-request aim to evenly balance replica load while prefix-cache and simple-session-affinity aim to maximize KV-cache reuse per request. NVIDIA Dynamo [NVIDIA, 2026b], SGLang Router [Zheng et al., 2024b], and Preble [Srivatsa et al., 2025] offer algorithms that jointly optimize for replica load and KV-cache locality.

GORGO takes inspiration from early content-delivery network design, which has many similarities to LLM serving. Wang et al. [2002] used three signals—server load, network proximity, and cache locality—to reduce response time and increase capacity. Evaluating network proximity can improve responsiveness, or TTFT, in geo-distributed serving environments. GORGO is the first LLM load balancing policy to also optimize (iii) network latency from client to server.

**Weight-selection Methods:** Methods for tuning the weights of a request scheduler fall into (i) offline and (ii) online protocols. Existing offline methods include Dynamo’s SLA Planner, which has a sophisticated method to autoscale prefill and decode workers to meet TTFT and ITL targets. It requires profiling before deployment to determine the optimal tensor parallelism configurations and scaling parameters for the planner to use during runtime [NVIDIA, 2026a].

GORGO uses online calibration because cache state, queue state, and batching behavior are intrinsically influenced by the routing policy’s decisions, and accurately replicating the conditions necessary to evaluate the policies requires a dedicated replay fleet, shadow execution, or a sufficiently accurate simulator of cache and continuous-batching dynamics. After all, a request log generated under one policy does not contain the counterfactual cache and queue evolution that would have arisen under another set of weights.

### 2.2 Cost Model

TTFT is known to consist of three different costs: network latency, prefill time, and queueing delay [He et al., 2025]. The KV-Cache, which stores key-value pairs of input sequences, removes redun-

dant prefill computation for user sequences sharing a prefix with previously computed sequences [Zheng et al., 2024b]. In a cross-region deployment setting, for an inference engine replica  $i$  holding a set of cached token prefixes  $c_i$ , the cost of TTFT can be defined as the following function, where  $x_r$  is the input sequence of tokens for a request  $r$  and the prefill time depends only on the set difference  $(x_r \setminus c_i)$ , the tokens in  $x_r$  not already cached on  $i$ :

$$\text{TTFT} = T_{\text{network}}(i) + T_{\text{queue}}(i) + T_{\text{prefill}}(x_r \setminus c_i) \quad (1)$$

Theoretically,  $T_{\text{network}}$  and  $T_{\text{queue}}$  correspond with round trip time from a client to server and the duration of request processing ahead of the incoming request  $r$ . However, inference engines support batching requests continuously in order to minimize waiting for completion of previous request processing [Yu et al., 2022]. While continuous batching allows admission of a new request into the currently running batch, the batch size is still bounded by a maximum number of concurrent requests and the available KV-cache budget, a critical knob that governs how much load a replica can admit before further requests wait in a queue [Kwon et al., 2023].

In a distributed system, the temporal delay of retrieving queueing metrics from an engine replica makes cost evaluation challenging. We represent the input to  $T_{\text{queue}}$  as the total number of tokens of requests without a completion event on the client proxy.  $T_{\text{network}}$  is measured trivially as the exponential weighted moving average of a ping’s round trip time from client proxy to server.

$$\text{TTFT} = T_{\text{network}}(\text{RTT}_i) + T_{\text{queue}}\left(i, \sum_{j=1}^{n_r} x_j\right) + T_{\text{prefill}}(x_r \setminus c_i) \quad (2)$$

### 2.3 GORGO Proxy Design

The GORGO proxy routes client requests to the engine replica with the minimum calculated cost from Equation 2, parameterized by weights  $W_{\text{rtt}}$ ,  $W_{\text{prefill}}$ , and  $W_{\text{queue}}$ . These parameters weight the inputs  $T_{\text{network}}$ ,  $T_{\text{queue}}$ , and  $T_{\text{prefill}}$ , which unfairly mixes the units of time and tokens, to normalize the correlated costs of latency, prefill cost, and queueing time on TTFT. The weight of  $T_{\text{prefill}}$  is fixed to 1 because GORGO proxy makes routing decisions based on relative replica cost: only the ratio of weights matters in this design.

$$\text{TTFT} = W_{\text{rtt}} * T_{\text{network}}(\text{RTT}_i) + W_{\text{queue}} * T_{\text{queue}}\left(i, \sum_{j=1}^{n_r} x_j\right) + T_{\text{prefill}}(x_r \setminus c_i) \quad (3)$$

GORGO proxy uses a simple (1+1) evolutionary strategy to tune weights  $W_{\text{rtt}}$  and  $W_{\text{queue}}$  on the objective function, p95 TTFT. Each parent weight  $x_{t,k}$  is perturbed multiplicatively in log-space by a normal random variable  $z_k$  times step size  $\sigma$ , and the new offspring weight  $x'_k$  is clamped to values in the hyperparameter range  $[lo_k, hi_k]$  where  $lo_k > 0$  and  $hi_k > 0$ .

$$x'_k = \text{clip}(\exp(\ln(x_{t,k}) + \sigma_t * z_k), [lo_k, hi_k]) \quad (4)$$

When  $x'$  beats the parent weight  $x_t$  on the objective metric, the incumbent weight  $x_{t+1}$  is updated to  $x'$ , and  $\sigma$  is adjusted to maintain Rechenberg’s 1/5 success rule [Rechenberg, 1973] of roughly one accepted offspring for every five proposals.

## 3 Dataset

Existing LLM chatbot datasets lack two critical components for benchmarking cache-aware policies: (i) prefill-bound requests with long-context prompts and (ii) multi-turn workloads with high prefix-reuse between requests. For example, we measure the average request length and global prefix reuse of **LMSYS-Chat-1M** [Zheng et al., 2024a] and **WildChat-4.8M** [Zhao et al., 2024], two popular LLM datasets derived from public chatbot demos (Table 1, Figure 1). WildChat-4.8M contains

Table 1: Dataset. ART-Chat-2.5M contains higher average input tokens and global prefix reuse, which is measured by adding the intra-user reuse and cross-user reuse, over other chat datasets.

| Dataset       | Total reqs | Users     | Avg input tokens | Intra-user reuse | Global reuse |
|---------------|------------|-----------|------------------|------------------|--------------|
| LMSYS-Chat-1M | 1,000,000  | —         | 467              | —                | 3.4%         |
| WildChat-4.8M | 3,199,860  | 1,833,730 | 2,925            | 4.7%             | 32.5%        |
| ART-Chat-2.5M | 2,525,215  | 4,984     | 17,964           | 89.4%            | 89.7%        |

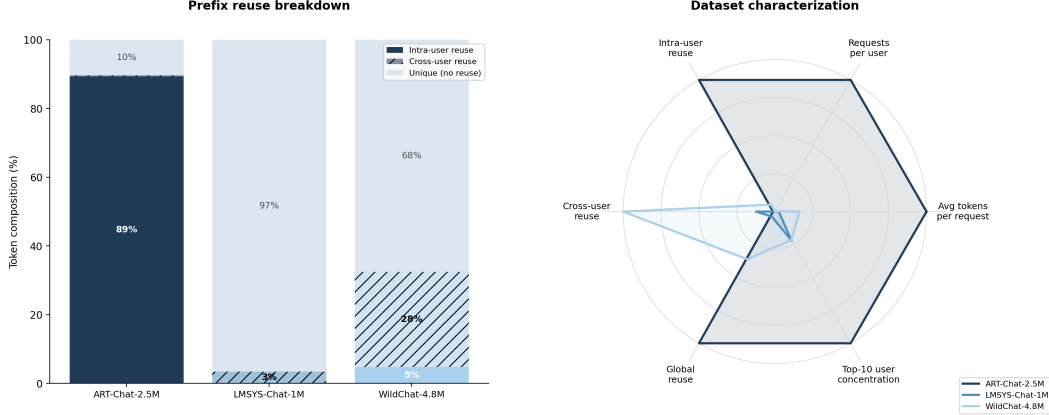


Figure 1: *Left*: Dataset characterization. ART-Chat-2.5M’s prefix reuse is overwhelmingly intra-user (89%) with minimal cross-user reuse (0.3%), which reflects the long-context, multi-turn data; WildChat’s prefix reuse is predominantly cross-user (28%, shared templates); LMSYS has minimal cross-user reuse (3%) and no intra-user reuse due to the lack of a field for client origin. *Right*: ART-Chat-2.5M contains the maximum for each attribute except for cross-user reuse, which in WildChat is a function of the shorter average token length (2,925 tokens) and a common system prompt.

hashed IPs per request, allowing categorization of cross-user and intra-user reuse while LMSYS-Chat-1M lacks user identification. Additional results from benchmarking GORGO on WildChat-4.8M can be found in Appendix C.

**ART-Chat-2.5M** is a long-context, multi-turn dataset synthetically generated from a week-long metadata trace of production inference traffic with the same prefix-reuse structure as the original workload. We release a replay-ready trace in the Mooncake FAST’25 format, which contains per-request timestamps, request metadata, and synthetically generated chat completion data [Qin et al., 2025]. By storing request timestamps, one can linearly scale the time between requests to control replica load. We characterize the dataset against WildChat-4.8M and LMSYS-Chat-1M in Table 1 and Figure 1. Notably, the intra-user prefix reuse and average token length in ART-Chat-2.5M are  $19\times$  and  $6\times$  higher than in WildChat-4.8M.

## 4 Experimental Setup

**Proxy and engine configuration.** Each policy runs the GORGO proxy on a small CPU worker in us-ashburn and controls a dedicated SGLang inference engine in each of the following regions: us-ashburn, eu-frankfurt, and ap-seoul. The engines contain two L40S GPUs each and serve the Qwen3.5-35B-A3B model in FP8 format [Qwen Team, 2025]. All policies are benchmarked on the same workload in parallel to rule out any variance in network conditions. The round-trip time between the us-ashburn proxy and engines during the tuning window is plotted in Figure 2. Due to the Qwen model’s limited context length of 32,768 tokens, we filter out any requests that contain  $>24,000$  tokens to leave adequate KV headroom. In SGLang, we set `max_concurrent_requests` to 64 and `max_output_tokens` to 128 to limit unnecessary decode while simulating adequate load on the replica. All workloads run alongside the proxy, dispatching requests to a local chat completion endpoint.

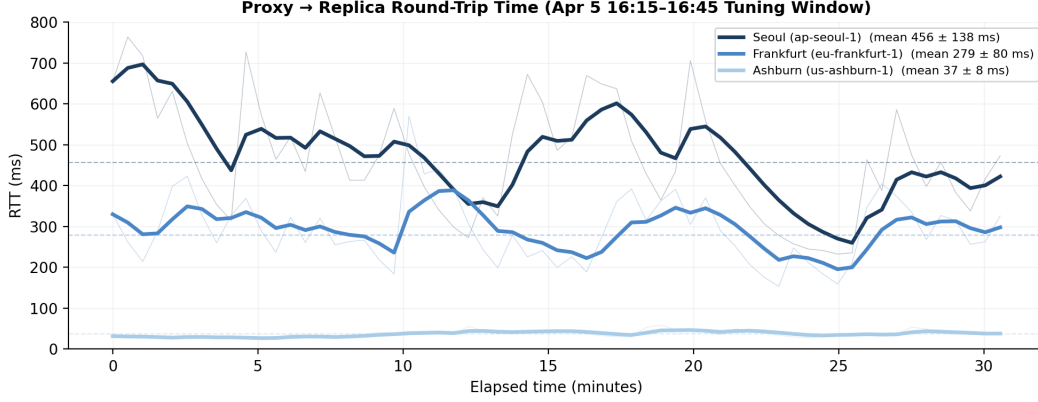


Figure 2: Round trip time (RTT) between the GORGO policy’s proxy in us-ashburn and the SGLang engines in us-ashburn, eu-frankfurt, and ap-seoul over the Apr 5 16:15–16:45 tuning window. The proxy measures RTT every 30 seconds on a separate probe from the `/metrics` request and smoothes the RTT with an exponential moving average (EWMA). Each region’s mean latency demonstrates that network latency either adds a consideration when choosing between replicas with disparate load and prefill cost or simplifies the choice when choosing between replicas with similar costs.

**Baseline policies.** We benchmark the GORGO policy and compare performance of both online and static modes to the below baselines. All SGLang metrics are scraped every 30 seconds from the engine’s Prometheus `/metrics` endpoint.

1. `least-load` minimizes the sum of proxy-tracked queued requests with SGLang metrics `num_running_reqs`, `num_queue_reqs`, and `num_used_tokens`. Queued requests to the proxy are defined as recently dispatched requests without a token response, and `num_used_tokens` are the currently occupied per-token KV slots.
2. `least-request` chooses the replica with the fewest in-flight requests from the proxy.
3. `prefix-cache` matches the request’s prefix to the replica with the highest prefix-cache overlap, tracked on the proxy-side by a prefix trie of dispatched requests [The AIBrix Team, 2025].
4. `simple-session-affinity` hashes the first 256 tokens from a request and routes to the replica with that prefix hash.

**Tuning and evaluation windows.** We pick three 30-minute windows with high user diversity from the ART-Chat-2.5M trace: Apr 5th 16:15–16:45, Apr 6 15:05–15:35, and Apr 7 19:45–20:15. Statistics on each of these windows are found in Table 5 (Appendix A). We assign Apr 5th as the window where we tune GORGO’s weights online to minimize the p95 TTFT of a rolling 128-request window with hop size 32.  $w_{rtt}$  and  $w_{queue}$  are each initialized to 0.5 and 0.1 and restricted to ranges  $[0.05, 2.0]$  and  $[0.05, 0.5]$ . These values are hand-picked from the paradigm of continuous batching in SGLang, where incoming requests can be scheduled into the current batch, affecting TTFT less significantly than a fixed floor of network latency between regions. The Apr 6–7 windows fix weight values GORGO learned on the Apr 5 tuning window. Due to the greater number of requests in Apr 6–7, we increase `time_scale` to 2.0 and 3.0, respectively, to control replica saturation (Table 4).

## 5 Results

**Results across three windows.** Table 2 reports TTFT, E2E latency, and inter-token latency (ITL) for all policies in all three windows. While the GORGO policy’s weights are updated to minimize p95 TTFT during tuning, the held-out, fixed-weight evaluation shows generalization of the learned values across days, with GORGO improving p95 TTFT by 6.9–15.5% and E2E latency by 14.3–30.9% over session-affinity. GORGO slightly underperforms baseline policies on the tuning window because the evolutionary strategy actively explores the space of parameters and tests worse weights than the learned solution, which converges after 672 samples (Figure 3).

Table 2: Experiment results. In the tuning window, GORGO learns weights  $w_{\text{rtt}}=0.276$ ,  $w_{\text{queue}}=0.5$  after initializing at weights  $w_{\text{rtt}}=0.5$ ,  $w_{\text{queue}}=0.1$ . On held out evaluation windows, GORGO’s weights are frozen, and the policy outperforms every baseline policy on p95 TTFT.

| Policy                                                                    | TTFT p50 (ms) | TTFT p95     | TTFT p99     | E2E p50 (s) | E2E p95     | E2E p99      | ITL (ms)    |
|---------------------------------------------------------------------------|---------------|--------------|--------------|-------------|-------------|--------------|-------------|
| <i>Tuning window — Apr 5 16:15–16:45 (time_scale=1.0, 7,195 requests)</i> |               |              |              |             |             |              |             |
| <b>GORGO</b>                                                              | <b>673</b>    | 2,514        | 4,473        | <b>3.04</b> | <b>8.91</b> | <b>17.54</b> | <b>17.6</b> |
| simple-session-affinity                                                   | 712           | <b>2,428</b> | 5,190        | 3.48        | 18.27       | 22.61        | 20.2        |
| least-load                                                                | 763           | 2,447        | <b>4,349</b> | 3.63        | 13.69       | 17.57        | 21.7        |
| least-request                                                             | 968           | 3,970        | 7,012        | 4.37        | 16.62       | 20.81        | 25.2        |
| prefix-cache                                                              | 1,477         | 6,784        | 9,613        | 13.14       | 22.63       | 26.81        | 82.3        |
| <i>Held-out eval — Apr 6 15:05–15:35 (time_scale=2.0, 7,630 requests)</i> |               |              |              |             |             |              |             |
| <b>GORGO</b>                                                              | <b>491</b>    | <b>1,584</b> | <b>2,222</b> | <b>1.80</b> | <b>3.28</b> | <b>4.37</b>  | <b>9.0</b>  |
| simple-session-affinity                                                   | 627           | 1,875        | 3,056        | 2.01        | 4.75        | 7.34         | 10.3        |
| least-load                                                                | 616           | 1,818        | 2,885        | 2.17        | 4.06        | 5.78         | 10.7        |
| least-request                                                             | 634           | 1,852        | 2,484        | 2.08        | 3.99        | 5.23         | 9.9         |
| prefix-cache                                                              | 574           | 1,798        | 2,750        | 2.07        | 4.95        | 10.34        | 10.6        |
| <i>Held-out eval — Apr 7 19:45–20:15 (time_scale=3.0, 8,663 requests)</i> |               |              |              |             |             |              |             |
| <b>GORGO</b>                                                              | <b>398</b>    | <b>1,417</b> | <b>2,061</b> | <b>1.74</b> | <b>3.36</b> | 6.81         | <b>9.3</b>  |
| simple-session-affinity                                                   | 457           | 1,522        | 2,402        | <b>1.74</b> | 3.92        | 7.12         | <b>9.3</b>  |
| least-load                                                                | 495           | 1,669        | 2,368        | 1.95        | 4.07        | <b>6.22</b>  | 10.0        |
| least-request                                                             | 527           | 1,759        | 2,578        | 1.98        | 4.44        | 7.85         | 10.0        |
| prefix-cache                                                              | 533           | 1,861        | 2,872        | 2.20        | 12.00       | 20.04        | 11.8        |

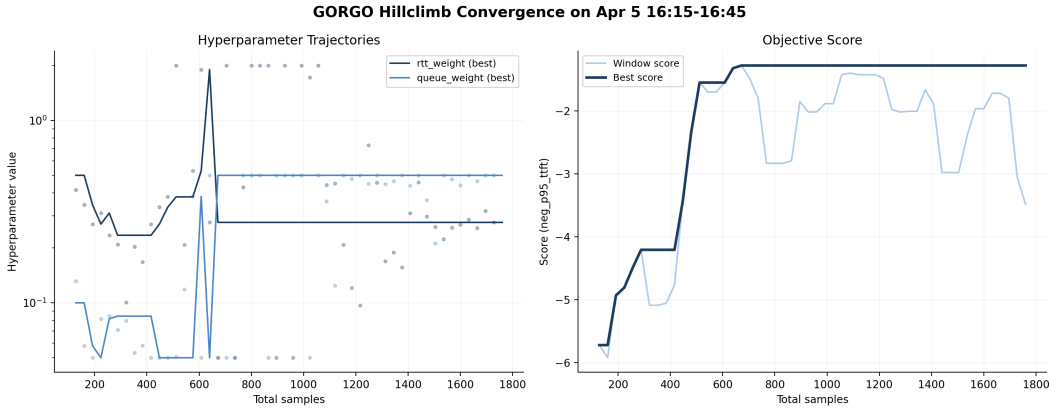


Figure 3: Convergence of the evolutionary strategy on GORGO policy weights  $w_{\text{queue}}$  and  $w_{\text{rtt}}$ . Both weights reach their local optima after 672 samples, which occurs on the 18th evolutionary step. The fitness function is the negative p95 TTFT (in seconds) of the rolling 128-request tuning window, so a smaller magnitude is better; the best score reached is  $-1.276$ , i.e. a 1.276 s best-window p95. Because it is computed over the rolling tuning window, this is lower than the 2,514 ms overall tuning-window p95 in Table 2.

Table 3: BFCL multi-turn replay. Bold marks the best value per column.

| Policy                  | TTFT p50 (ms) | TTFT p95     | TTFT p99     | E2E p50 (s) | E2E p95     | E2E p99     | ITL (ms) |
|-------------------------|---------------|--------------|--------------|-------------|-------------|-------------|----------|
| <b>GORGO</b>            | <b>3,328</b>  | <b>4,700</b> | 5,720        | <b>3.66</b> | <b>5.17</b> | <b>6.15</b> | —        |
| simple-session-affinity | 3,655         | 5,087        | <b>5,608</b> | 4.11        | 5.61        | 6.17        | —        |

**BFCL multi-turn replay.** As an additional high-prefix-reuse workload, we replay 200 teacher-forced episodes from BFCL’s `multi_turn_base` subset, where each episode contains sequential tool calls [Patil et al., 2025]. We use two Qwen3-4B H200 replicas in us-east and us-west, 128 concurrent episodes, and at most 256 output tokens per request. Identical prompts and tool results are replayed under GORGO and session affinity.

Under high load, GORGO reduces p50 and p95 TTFT by 8.9% and 7.6%, and p50 and p95 E2E latency by 10.8% and 7.9%, respectively.

**Load Sweep** We find when sweeping across the `time_scale` parameter that our SGLang inference engines reach a saturation point. After a concurrency threshold is reached, requests begin to experience HOL queueing delay. In Table 4, the policies experience a  $3\times$  improvement in p95 TTFT and p95 E2E latency after `time_scale=3.0` is reached. In the tuning window, most policies other than GORGO are over saturating replicas due to the abnormally high p95 E2E latency when compared to the p95 E2E latency in later windows. We recommend sweeping across timescales to find a clean under-saturated traffic profile before running GORGO. Due to the increased load profile of Apr 6-7, the timescale was increased to create a fair evaluation environment. However, we include in Appendix B a reference run of Apr 7 with `time_scale=2.0` to show saturation at a lower timescale.

Table 4: Load sweep. A sweep of the timescale variable, which linearly scales the time between requests, shows p95 TTFT and E2E latency degrading at lower timescales. After `time_scale=3.0`, metrics improve by over  $3\times$  for unsaturated policies. Notably, this timescale sweep shows that GORGO breaks the saturation point earlier than other policies due to a well-tuned load term.

| Policy                            | Input (tok/s) | TTFT p95 (ms) | E2E p95 (s) | Saturated? |
|-----------------------------------|---------------|---------------|-------------|------------|
| <i>time_scale=1.0 (full load)</i> |               |               |             |            |
| gorgo                             | 34,064        | 8,260         | 15.76       | yes        |
| simple-session-affinity           | 34,035        | 1,835         | 17.94       | yes        |
| least-request                     | 34,030        | 6,138         | 18.62       | yes        |
| <i>time_scale=2.0</i>             |               |               |             |            |
| gorgo                             | 17,003        | 1,822         | 5.65        | no         |
| simple-session-affinity           | 16,999        | 1,707         | 15.50       | yes        |
| least-request                     | 16,999        | 4,361         | 16.85       | yes        |
| <i>time_scale=3.0</i>             |               |               |             |            |
| gorgo                             | 11,327        | 1,378         | 3.25        | no         |
| simple-session-affinity           | 11,326        | 1,556         | 4.04        | no         |
| least-request                     | 11,326        | 1,805         | 4.92        | no         |

**Exploiting Continuous Batching in SGLang** In the continuous batching paradigm, we learned that GORGO can stumble upon abnormal weight values and achieve an impressively low p95 TTFT at the cost of high p95 E2E latency. Appendix D presents results from an experiment where the evolutionary strategy learns to set  $w_{\text{queue}}$  to 0 while keeping  $w_{\text{rtt}}$  near 0.23, which results in p95 TTFT 17% better than the next-best policy. For this window, GORGO chose to send 100% of requests to the closest replica in us-ashburn. Continuous batching [Yu et al., 2022] works by admitting incoming requests into the currently running batch as long as some concurrency slot exists for the request. Thus, when all requests are sent to the same replica, any new requests sent to that replica can enter the batch, reuse existing KV-cache, and prefill a single token. However, once the request enters the memory-bound decode phase, the replica’s memory headroom saturates, KV-cache thrashes between GPU and CPU memory, and ITL/E2E latency collapses [Yu et al., 2022, Kwon et al., 2023].

## 6 Related Work

**Prefix caches and reuse.** RadixAttention in SGLang [Zheng et al., 2024b] stores per-request KV state in a radix tree; PagedAttention in vLLM [Kwon et al., 2023] enables prefix sharing through paged memory. KVLink [Yang et al., 2025b], ChunkKV [Liu et al., 2025], KVFlow [Wang et al., 2025], and Learned Prefix Caching [Yang et al., 2025a] improve the single-replica cache but do not decide which replica serves a request.

**Cross-replica routing and cross-region serving.** Preble [Srivatsa et al., 2025] introduces longest-prefix-match routing across replicas with a load-balance fallback; AIBrix [The AIBrix Team, 2025] packages a production-grade variant of the same design and supplies the baseline we run. Mooncake [Qin et al., 2025] is a KV-cache-centric architecture and the source of our trace format. SkyServe [Mao et al., 2025] and SkyWalker [Xia et al., 2025] address cross-region LLM serving at the placement and spillover layers respectively, but treat per-request routing as out of scope. k-LPM [Dexter et al., 2025] formulates LLM scheduling under TTFT constraints as NP-hard. DLPM [Cao et al., 2025] targets fairness with locality. Llumnix [Sun et al., 2024] migrates requests across replicas. CacheBlend [Yao et al., 2025] fuses cached KV state from reused non-prefix chunks for retrieval-augmented generation, but does not address cross-replica routing or network RTT. DistServe [Zhong et al., 2024] and Splitwise [Patel et al., 2024] disaggregate prefill from decode. GORGO is the first single-router policy in this space that scores cache locality, replica load, and wide-area latency in one cost and derives the cost weights from the deployment’s own per-request TTFT stream, with no engine modifications.

**Online hyperparameter adaptation.** Unlike Bayesian-optimization or bandit approaches that maintain a surrogate model, the (1+1)-ES [Rechenberg, 1973] provides fast, overhead-free convergence in our 2-dimensional weight space.

## 7 Limitations

Several constraints bound our conclusions. First, GORGO does not currently model heterogeneous hardware setups, where replicas may have different prefill rates, queueing behavior, or KV-cache capacities. Second, because the online tuner optimizes p95 TTFT, it can exploit continuous batching by concentrating load on the nearest replica, trading E2E and ITL tails for TTFT (§5). Adding a lower bound to the queue term mitigates but does not eliminate this trade-off. Third, we do not consider prefill/decode disaggregation, which introduces KV-transfer cost. Finally, GORGO does not manage or directly observe replica-side KV-cache eviction. Consequently, the proxy’s cache index may temporarily overestimate prefix availability when a cached prefix is evicted between routing and execution.

## 8 Conclusion

Routing across LLM replicas is a three-signal decision balancing cache locality, replica load, and wide-area latency, but production heuristics commit to one signal and degrade when that stops being the limiting factor. We treat the three as terms of an additive per-replica cost and let online TTFT feedback optimize scaling weights, in place of operator-tuned constants or offline profiling. On a long-context, high-prefix-reuse production trace the GORGO policy family collectively achieves the lowest TTFT at every reported percentile across both held-out evaluation windows. On a held-out WildChat window, fixed-weight GORGO improves p95 TTFT by 17.2% and p99 TTFT by 37.6% over random routing (Appendix C). On paired BFCL multi-turn replays, it reduces p95 TTFT and E2E latency by 7.6% and 7.9% relative to session affinity. The design choices of GORGO transfer to deployments where the workload regime is not known in advance and a dedicated calibration window before each redeploy is not affordable, making it effective in production LLM serving on long context, distributed workloads.

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## A Evaluation Window Characteristics

Table 5: Workload characteristics of the three tuning and held-out windows in Table 2.

|                        | Apr5 tuning<br>16:15–16:45<br>time_scale 1.0 | Apr6 eval<br>15:05–15:35<br>time_scale 2.0 | Apr7 eval<br>19:45–20:15<br>time_scale 3.0 |
|------------------------|----------------------------------------------|--------------------------------------------|--------------------------------------------|
| Distinct users         | 579                                          | 606                                        | 540                                        |
| Requests / user        | 12.8                                         | 12.9                                       | 16.4                                       |
| Avg turns / conv.      | 30.3                                         | 27.7                                       | 21.5                                       |
| Multi-turn requests    | 73.9%                                        | 74.1%                                      | 73.7%                                      |
| Median input tokens    | 5,638                                        | 5,787                                      | 5,745                                      |
| Avg input tokens       | 6,877                                        | 7,008                                      | 7,148                                      |
| p95 input tokens       | 19,886                                       | 20,974                                     | 20,435                                     |
| Block global reuse     | 78.8%                                        | 77.6%                                      | 87.9%                                      |
| Block intra-user reuse | 78.5%                                        | 77.1%                                      | 87.4%                                      |

## B Apr 7 at time\_scale=2.0

Table 6: Apr 7 19:45–20:15 held-out window at time\_scale=2.0

| Policy                  | TTFT p50 (ms) | TTFT p95     | TTFT p99     | E2E p50 (s) | E2E p95     | E2E p99      | ITL (ms)    |
|-------------------------|---------------|--------------|--------------|-------------|-------------|--------------|-------------|
| GORGO                   | 586           | 1,891        | 3,431        | 2.38        | <b>7.10</b> | <b>12.58</b> | 13.4        |
| simple-session-affinity | <b>512</b>    | <b>1,684</b> | <b>2,455</b> | <b>2.19</b> | 14.66       | 17.19        | <b>12.2</b> |
| least-load              | 633           | 2,158        | 3,753        | 2.69        | 10.39       | 16.19        | 15.5        |
| least-request           | 915           | 4,806        | 7,114        | 3.77        | 17.27       | 19.61        | 19.7        |
| prefix-cache            | 1,474         | 8,243        | 12,167       | 7.67        | 22.14       | 26.66        | 45.7        |

## C WildChat replay

Table 7: Held-out WildChat-4.8M replay at time\_scale=2.0 (5.7 requests/s open-loop),  $c=64$ , using one H100 replica in each of us-west4, CANADA-2, and sines-2. Fixed-weight GORGO uses the weights learned on the first window ( $w_{\text{rtt}}=2.0$  and  $w_{\text{queue}}=0.104$ ). Each arm contains 20,422 completed requests with zero failures. Bold marks the best value per column; ITL is the median.

| Policy | TTFT p50 (ms) | TTFT p95   | TTFT p99   | E2E p50 (s) | E2E p95     | E2E p99     | ITL (ms)   |
|--------|---------------|------------|------------|-------------|-------------|-------------|------------|
| GORGO  | <b>215</b>    | <b>506</b> | <b>728</b> | 1.55        | <b>2.42</b> | <b>2.98</b> | 10.1       |
| random | 244           | 611        | 1,166      | <b>1.38</b> | 2.51        | 4.04        | <b>8.4</b> |

## D Load Weight Adaptation

Table 8: Reward hacking on the Apr 2 12:30–13:00 window. With  $w_{\text{queue}}=0$ , the tuned gorgo-static wins every TTFT percentile but is worst on the E2E and ITL tails, because it routes  $\sim 100\%$  of traffic to a single replica that saturates under load. Bold marks the best value per column; ITL is the median.

| Policy                                                                    | TTFT p50 (ms) | TTFT p95     | TTFT p99     | E2E p50 (s) | E2E p95     | E2E p99     | ITL (ms)   |
|---------------------------------------------------------------------------|---------------|--------------|--------------|-------------|-------------|-------------|------------|
| <i>Held-out eval — Apr 2 12:30–13:00 (time_scale=1.0, 3,071 requests)</i> |               |              |              |             |             |             |            |
| gorgo-static (hacked)                                                     | <b>180</b>    | <b>1,072</b> | <b>1,810</b> | 2.05        | 12.49       | 16.63       | 14.1       |
| simple-session-affinity                                                   | 268           | 1,715        | 2,947        | 1.74        | 3.88        | 5.90        | 10.8       |
| least-load                                                                | 278           | 1,669        | 2,665        | 1.69        | 3.89        | 8.94        | 10.1       |
| least-request                                                             | 274           | 1,285        | 1,902        | <b>1.37</b> | <b>2.61</b> | <b>3.34</b> | 8.6        |
| random                                                                    | 283           | 1,456        | 2,124        | 1.40        | 2.92        | 4.24        | <b>8.5</b> |
| prefix-cache                                                              | 288           | 1,468        | 2,374        | 1.56        | 3.59        | 7.12        | 9.4        |

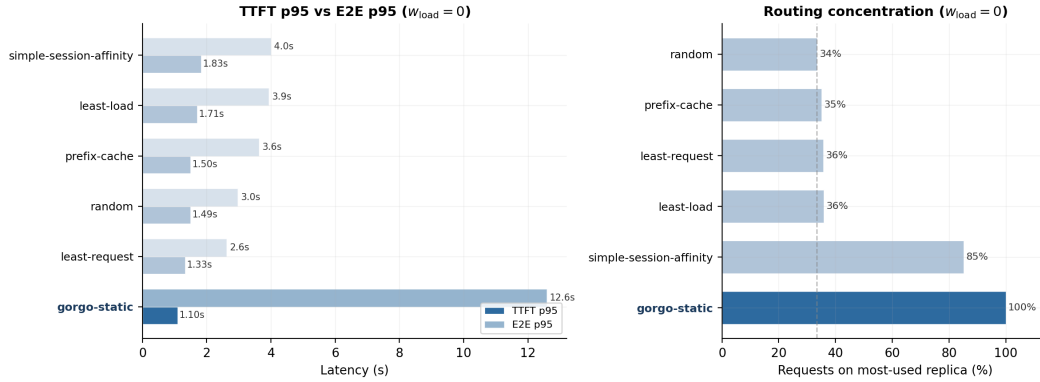


Figure 4: *Left*: TTFT p95 (dark) vs. E2E p95 (light) on the midday diurnal trace with  $w_{\text{queue}}=0$ . gorgo-static wins TTFT but its E2E inflates to 12.6s. *Right*: routing concentration. gorgo-static sends 100% of requests to one replica.